ABDULLAH SÖNMEZ

Electrical & Electronics Engineer (B.Sc., 100% English Instruction) • Mechatronics A.S. (Graduated 3rd in Department)

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Technical Focus: Industrial Power Electronics & Induction Heating Hardware • IGBT Gate Drivers & Oscilloscope Diagnostics • Multi-Rail DC-DC & Sensor PCB Design (KiCad) • Model-Based HDL Workflow (Simulink / Vivado)

1. Education & Upper-Division Academic Trajectory

Degree & Academic Program	Institution & Location	Academic Distinctions & Key Metrics	Graduation
B.Sc. in Electrical & Electronics Engineering (100% English Medium of Instruction)	**Dokuz Eylül University** Faculty of Engineering, Izmir, Türkiye	**Design & Research Coursework GPA: 3.88 / 4.00** Cumulative Grade: **71.00 / 100** (2.10/4.00 DEU scale) English Preparatory Program Score: **85 / 100**	Jul 2025
A.S. (Associate Degree) in Mechatronics	**Necmettin Erbakan University** Konya, Türkiye	**Graduated 3rd in Department (High Honors)** Hands-on electromechanical systems, sensors & industrial automation	Jul 2018
Technical High School Diploma (Computer Programming Track)	**Kılıçarslan İMKB Teknik Anadolu Lisesi** Konya, Türkiye	Foundational C/C++ programming, digital logic & computer hardware	Jun 2016

Upper-Division Project, Research & Core Coursework Distinction (Dokuz Eylül University): While early theoretical courses impacted the overall 4.0-scale average (under DEU's strict scale where 70–74 = 2.00 and 60–64 = 1.00), performance across all capstone, directed research, hardware design, and core technical courses demonstrated strong upward momentum:

Course Code & Title	Course Category	Term	ECTS	Grade
EED 4894 — Final Year Project (Model-Based FPGA Workflow Thesis)	Undergraduate Thesis	Sem 8 (2025)	10.0	**AA (4.0)**
EED 4795 — Research Project & EED 3009 — Engineering Design II	Research & Hardware Design	Sem 7 & Sem 5	6.0	**AA (4.0)**
MTH 3502 — Information Networks & PHY 1101 — Physics I	Technical & Science Core	Sem 6 & Sem 1	7.0	**AA (4.0)**

Course Code & Title	Course Category	Term	ECTS	Grade
EED 1014 — Engineering Design I & EED 3507 — Electromagnetic Wave Theory	Design & Electromagnetics	Sem 2 & Sem 5	7.0	**BA (3.5)**
EED 4401 — Biomedical Engineering Fundamentals	Technical Elective	Sem 7 (2024)	3.0	**BA (3.5)**
EED 3018 — Microprocessor Systems & EED 2301 — Circuit Theory I	Embedded & Circuit Core	Sem 6 & Sem 3	11.0	**BB (3.0)**
EED 4502 — Analog Integrated Circuits & EED 3020 — Energy Conversion II	IC & Power Machinery Core	Sem 8 & Sem 6	12.0	**CB (2.5)**

2. Professional Engineering, Military Service & Industrial Experience

Reserve Officer (Second Lieutenant / Engineering Officer) — Turkish Armed Forces | Aug 2026 – Present (Türkiye)

- **Technical Operations & Leadership Service:** Currently fulfilling mandatory national service as a commissioned Reserve Officer (*Yedek Subay*), managing technical equipment readiness, disciplined team coordination, and operational maintenance workflows (scheduled completion prior to Fall 2027 graduate enrollment).

Electrical-Electronics Engineer — Mahes Machine (Induction Heating Systems) | May 2026 – Jul 2026 (Konya, Türkiye)

- **High-Power Induction Heating Assembly & Diagnostics:** Participated in the electromechanical cabinet assembly, water-cooled copper busbar integration, and functional testing of industrial high-frequency induction heating machines prior to military service.
- **IGBT Gate-Drive Testing & Oscilloscope Verification:** Conducted component-level fault diagnosis, blown 600A dual IGBT module (`Fuji 2MBI600`) replacement, and gate-emitter switching waveform checks at **27.72 kHz** using digital (`Tektronix TDS 210`) and analog (`MOS-620 20MHz`) oscilloscopes on hybrid isolated gate driver boards (`MHS-PCB16` with `VLA567-11R` and `1.5KE` TVS protection).
- **Control, Trigger & Isolated SMPS Board Assembly:** Performed dip/wave solder-pot assembly, conformal coating, and component replacement on frequency/trigger boards (`MHS-PCB19.R1` with `TL084`, `CD4081`, `CD4093`, `CD40106`, and optocoupler interlocks), VCO display cards (`LM566`, `CA3140`, `LM324`, `LM393`), and multi-output isolated flyback SMPS boards.

Production & Commissioning Engineer — CW Enerji (Photovoltaic Cell Manufacturing) | Jul 2025 – Aug 2025 (Antalya, Türkiye)

- **PV Solar Cell Line Validation:** Contributed to machinery commissioning, parameter calibration, and validation processes during the installation and revision phase of a photovoltaic (PV) solar cell production line; prepared short-term efficiency reports for maintenance teams.

Electrical-Electronics Engineering Intern — Bilimsel Akademi | Jul 2024 – Sep 2024 (Izmir, Türkiye)

- **C/C++ Programming & Software Fundamentals:** Practiced C and C++ algorithm development, pointer arithmetic, Object-Oriented Programming (OOP) data structures, and version control/debugging workflows during undergraduate summer internship training.

3. Core Technical Competencies & Laboratory Instrumentation

Engineering Domain	Hands-On Hardware, Integrated Circuits & Software Tools
Power Electronics & Industrial Hardware	Industrial Induction Heating Systems, 600A Dual IGBT Modules (`Fuji 2MBI600`), Hybrid Isolated Gate Drivers (`VLA567-11R`), TVS Snubber Networks (`1.5KE`), Multi-Output Isolated Flyback SMPS, Synchronous Buck DC-DC (`TPS54302`), USB-PD 20V Sink (`CH224K`)
Analog/Digital Control & Board Testing	Op-Amp & Comparator Conditioning (`TL084`, `LM324`, `CA3140`, `LM393`), Voltage-Controlled Oscillators (`LM566 VCO`), CMOS Schmitt Trigger Logic (`CD4081`, `CD4093`, `CD40106`), Optocoupler Fault Interlocks, Linear Regulators (`7815/7915`, `AMS1117-3.3`)
PCB Layout & Lab Instrumentation	KiCad (Schematic Capture, 2-Layer Copper Polygon-Pour PCB Layout, RF Keep-Out Routing, DRC/ERC Verification, 3D Viewer, Gerber/BOM), Digital & Analog Oscilloscopes (`Tektronix TDS 210`, `MOS-620`), Multimeters, Bench Power Supplies, SMD/THT & Solder-Pot Rework
Simulation, Modeling & Programming	MATLAB/Simulink (Model-Based Design, Vision HDL Toolbox, HDL Coder Workflow), Xilinx Vivado 2024.2 (RTL Synthesis & Behavioral Simulation Flow), C/C++, Python (Basic), ESP32 / Microcontroller Interfacing (I2C, UART, Relay Actuation)

4. Project I: Model-Based HDL Workflow Evaluation for Image Enhancement (B.Sc. Thesis)

B.Sc. Capstone Thesis (EED 4894, Grade: AA, June 2025) — *Advisor: Prof. Dr. Uğur Çam (Committee: Prof. Dr. Özge Cihanbeğendi, Prof. Dr. Taner Abdullah Oğuzer, Chairman: Prof. Dr. Yavuz Şenol)*. Archive: 01-FPGA-Image-Enhancement-Vivado.

- **Model-Based Design & Reference Architecture Study:** Investigated the model-based FPGA design methodology in MATLAB/Simulink by configuring and analyzing MathWorks Vision HDL Toolbox reference architectures for real-time **Global Histogram Equalization (HE)** (4-part spatial video partitioning and 256-bin accumulation) and **Contrast Limited Adaptive Histogram Equalization (CLAHE)** (8×8 tile division, clip-limit contrast limiting, and interpolation).
- **Automated HDL Code Generation & Vivado Synthesis Flow:** Executed the automated **MATLAB HDL Coder** workflow (running pre-codegen compatibility checks via the HDL Workflow Advisor to generate SystemVerilog source and testbench files) and imported the generated code into **Xilinx Vivado 2024.2** to inspect synthesized RTL schematics, hardware resource utilization reports (`Kintex-7` / `Spartan-7`), and behavioral simulation waveforms.

Evaluated Reference Model	Algorithmic Structure in Simulink	HDL Coder Workflow Step	Vivado 2024.2 Inspection & Simulation
Global Histogram Equalization (HE)	4-way spatial video partition (240×320, 8-bit, 256 histogram bins)	HDL Workflow Advisor validation & automated SystemVerilog codegen	RTL schematic generation, BRAM utilization review & testbench waveform check

Evaluated Reference Model	Algorithmic Structure in Simulink	HDL Coder Workflow Step	Vivado 2024.2 Inspection & Simulation
Contrast Limited Adaptive HE (CLAHE)	8×8 local tiles (30×40 px/tile) with clip limit & tile interpolation	HDL Workflow Advisor validation & automated SystemVerilog codegen	RTL subsystem inspection, DSP/BRAM synthesis report & behavioral simulation

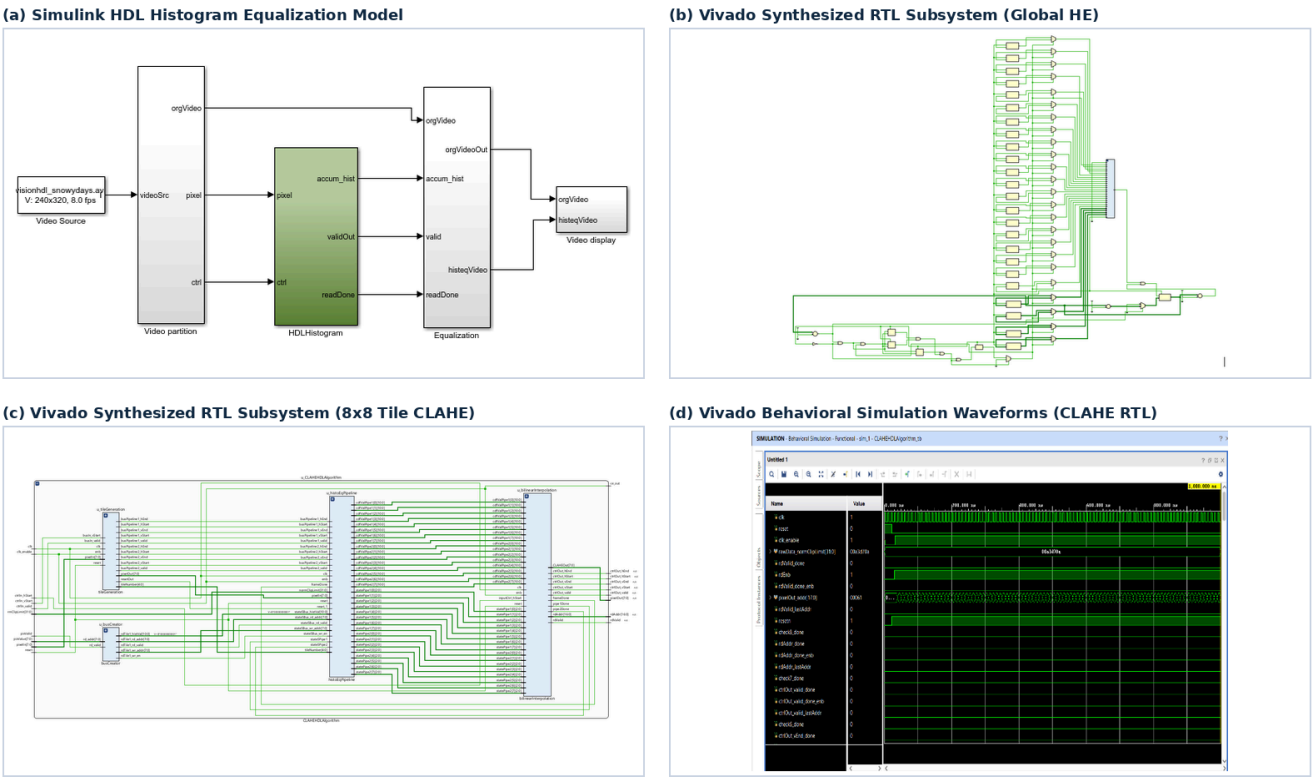


Figure 1: Model-based HDL Coder to Vivado workflow from B.Sc. Thesis: (a) Simulink Vision HDL Histogram Equalization model with 4-way video partitioning; (b) Xilinx Vivado 2024.2 synthesized RTL schematic of the generated `HDLHistogram` module; (c) Vivado synthesized RTL view of the generated 8×8 tile CLAHE subsystem; (d) Behavioral simulation waveforms in Xilinx Vivado verifying testbench execution.

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5. Project II: 100W USB-C PD 20V Sink & Multi-Rail Buck Power Distribution Board (KiCad)

Multi-Rail DC-DC Power Supply Schematic & PCB Layout Practice (`100W_USBC_PD_SMPS` — V1.0) — Archive: [02-100W-USBC-PD-MultiRail-SMPS-KiCad]:

- **USB-C Power Delivery & Step-Down Regulator Topology:** Designed a four-output DC-DC power distribution board around a 16-pin USB-C receptacle (J1) and a CH224K USB-PD sink controller (U1) configured via CFG1–CFG3 to request a +20V DC bus from a PD adapter (providing direct +20V passthrough on terminal J2). Stepped down the +20V rail using two TPS54302 3A synchronous buck regulator ICs—U4 with a 10µH inductor (L2) and 100kΩ/5.23kΩ voltage divider ($V_{out} = 0.596 \times (1 + 100/5.23) \approx 12\text{V}$) on J5, and U2 with a 10µH inductor (L1) and 100kΩ/13.3kΩ divider ($\approx 5\text{V}$) on J4—followed by an AMS1117-3.3 linear regulator (U3) providing **+3.3V** on J3.
- **2-Layer Copper Polygon-Pour PCB Layout & DRC/ERC Checks:** Completed the 2-layer PCB layout in KiCad using wide copper polygon pours for the +20V input and regulated output rails, placing decoupling

and bootstrap capacitors (100nF boot, 10µF input, 22µF output) adjacent to the IC pins, and verifying the layout with **0 DRC errors** and **0 ERC errors**.

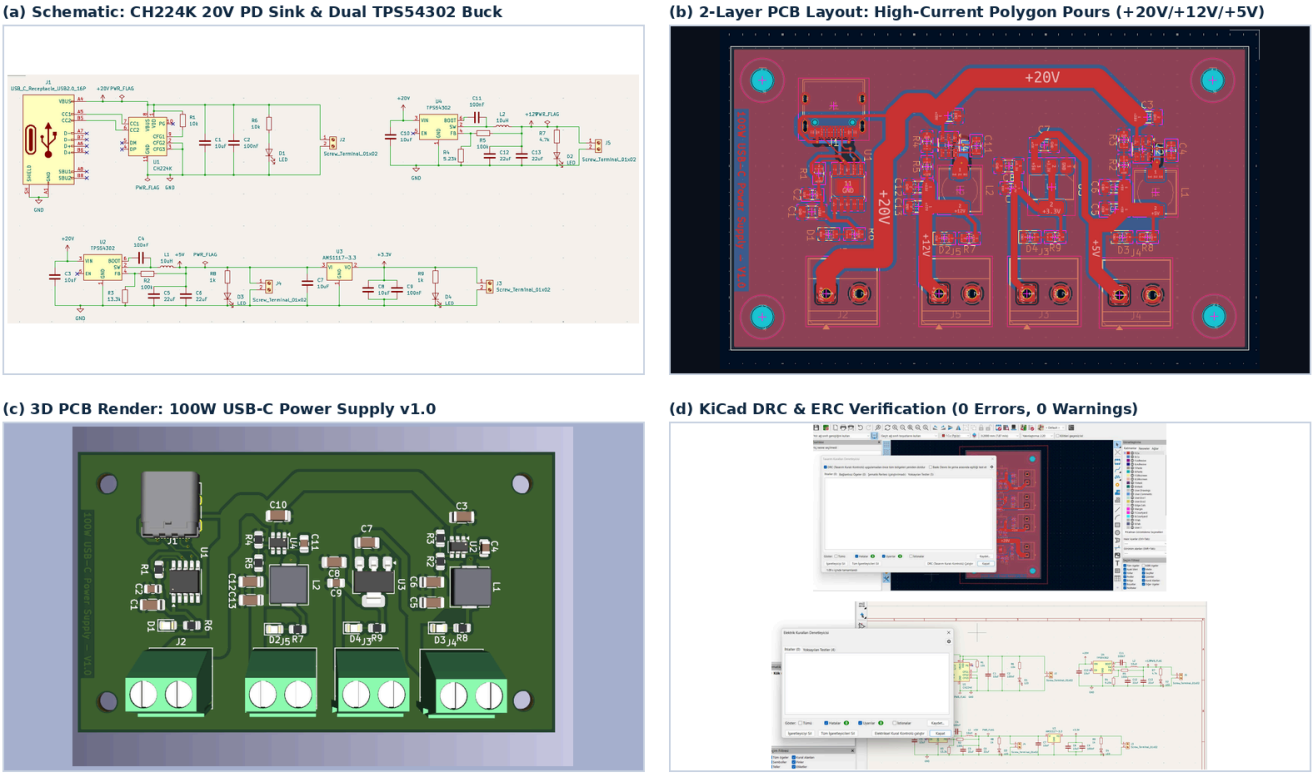


Figure 2: KiCad design and verification visuals for `100W_USBC_PD_SMPS` (V1.0): (a) Schematic combining `CH224K` 20V USB-PD sink negotiation, dual `TPS54302` buck stages (+12V and +5V), and an `AMS1117-3.3` LDO; (b) 2-layer PCB layout with wide copper zones for `+20V`, `+12V`, `+5V`, and `+3.3V`; (c) 3D PCB viewer model; (d) KiCad DRC and ERC verification logs confirming 0 errors and 0 warnings.

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6. Project III: ESP32-WROOM-32 Smart Sensor & Relay Gateway PCB (KiCad)

Embedded IoT Hardware Schematic & 2-Layer PCB Layout Practice (`Smart_Sensor_Gateway_Rev1` — V1.0) — Archive: 03-ESP32-Smart-Sensor-Gateway-KiCad:

- **USB-C Power Input, Protection & LDO Regulation:** Designed a 2-layer sensor and relay interface board around the `ESP32-WROOM-32` Wi-Fi/BLE module (U2). Configured the 16-pin USB-C input (`J1`) with two `5.1kΩ` pull-down resistors (`R1, R2`) on the `CC1/CC2` pins to enable 5V VBUS delivery from USB-C hosts, protected by a TVS diode (D1), and regulated down to +3.3V via an `AMS1117-3.3` LDO (U1) with `10µF` and `100nF` decoupling capacitors.
- **Sensor Expansion Headers, Relay Driver & RF Keep-Out Layout:** Included two 4-pin sensor headers (`J2, J3`) with `4.7kΩ` I2C pull-up resistors (`R7, R8`), hardware reset/boot pushbuttons (`SW1, SW2`), and a low-side NPN transistor switch (Q1 with 1kΩ base resistor R9) driving a 5V Form-C electromechanical relay (K2, `SANYOU SRD`) with a `1N4007` flyback diode (D4) to clamp inductive kickback. Routed the 2-layer PCB with a copper-free keep-out zone under the ESP32 antenna and thickened high-current traces for the relay screw terminal (`J4`), passing KiCad **DRC and ERC with 0 errors**.

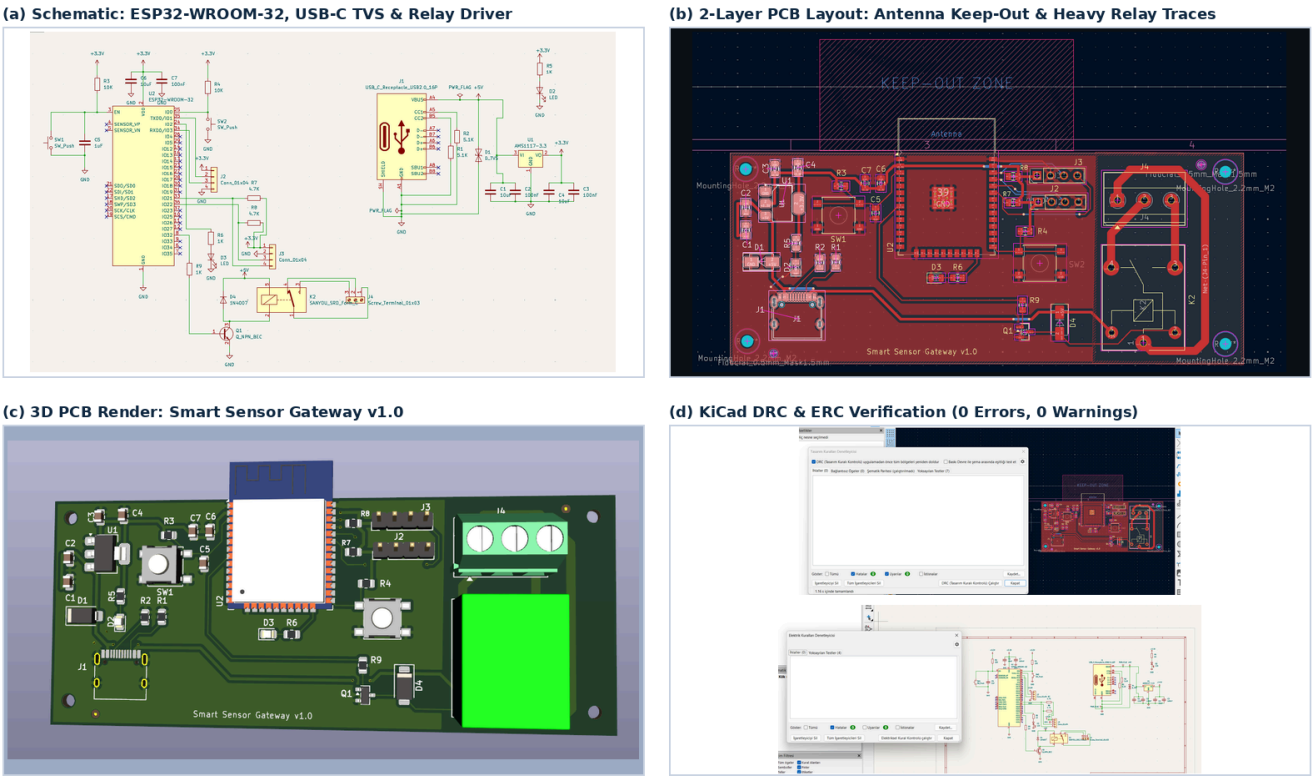


Figure 3: KiCad design and verification visuals for `Smart_Sensor_Gateway_Rev1` (V1.0): (a) Schematic featuring `ESP32-WROOM-32`, USB-C 5.1k CC pull-downs and TVS diode, `AMS1117-3.3` LDO, I2C/UART headers, and NPN relay driver; (b) 2-layer PCB layout showing the RF antenna copper keep-out zone and widened relay contact traces; (c) 3D PCB render; (d) KiCad DRC and ERC verification logs confirming 0 errors and 0 warnings.

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7. Project IV: Industrial Resonant Induction Heating Hardware & Diagnostics (Mahes Machine)

7.1 Isolated IGBT Gate-Drive Diagnostics, 600A Module Replacement & Oscilloscope Verification

Hands-on assembly, component-level troubleshooting, and oscilloscope testing across Mahes Machine's industrial induction heating power stages (04-Industrial-Induction-Heating-Power-Electronics):

- **Isolated IGBT Gate Driver (`MHS-PCB16`) & 600A Module Diagnostics:** Assembled, serviced, and tested gate driver cards built around the **`VLA567-11R` hybrid isolated gate driver IC**, electrolytic decoupling capacitor banks, and `1.5KE` bidirectional TVS protection diodes. Replaced damaged 600A dual IGBT modules (`Fuji 2MBI600`) and verified clean **27.72 kHz** gate-emitter switching square waves (`5V/div, 50µs/div`) using a `Tektronix TDS 210` digital oscilloscope and a `MOS-620 20MHz` analog oscilloscope.
- **Resonant Control & VCO Trigger Architecture (`MHS-PCB19.R1`):** Documented and tested analog/CMOS control boards comprising `LM566` voltage-controlled oscillators (VCO), `CA3140` BiMOS op-amps, `TL084` quad JFET op-amps, `LM324/LM393` comparators, `CD4081/CD4093/CD40106` Schmitt trigger logic gates, and optocoupler-isolated fault trip channels (`IGBT_ALARM`, `SU_ALARM` water-flow switch, and `ISI_ALARM` thermal thermostat interlocks).

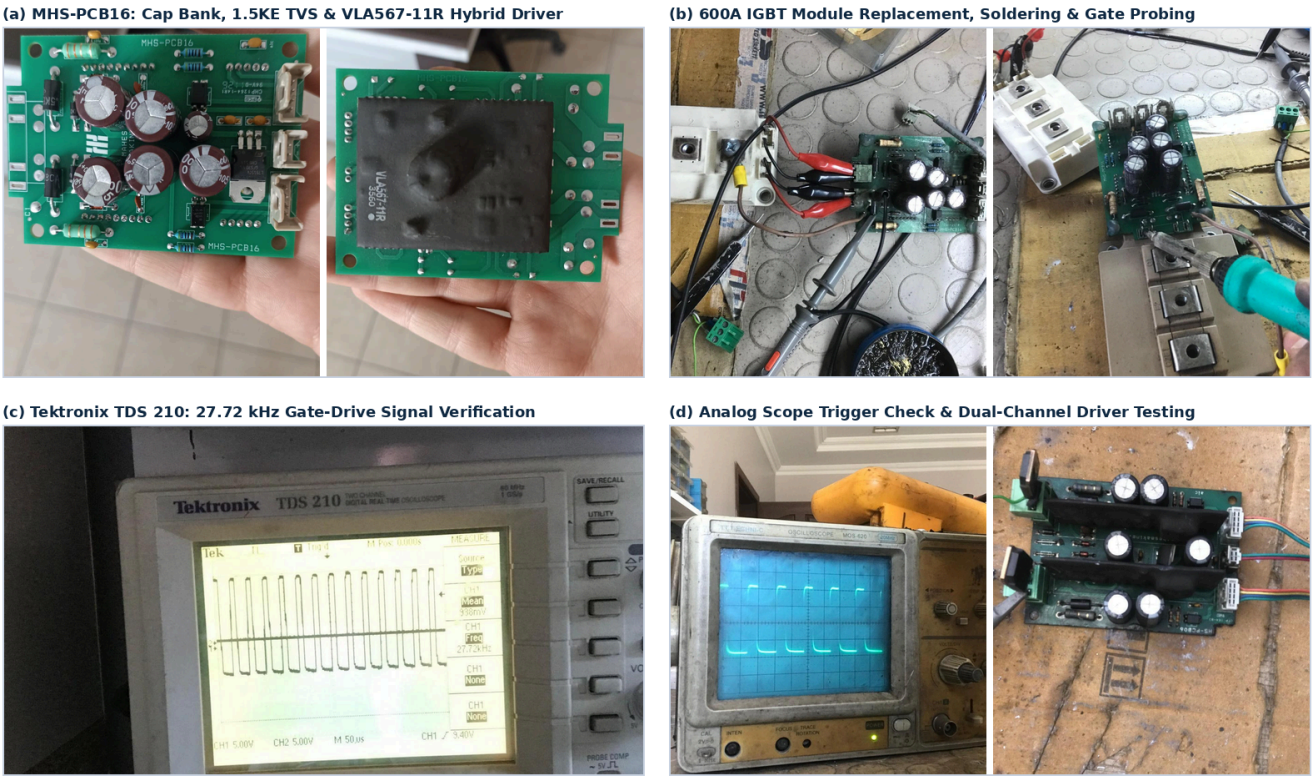


Figure 4: Board-level IGBT gate driver diagnostics and oscilloscope verification at Mahes Machine: (a) Top and bottom views of the `MHS-PCB16` gate driver and snubber card highlighting the `VLA567-11R` hybrid isolated driver IC and `1.5KE` TVS diodes; (b) Bench probing and gate-terminal soldering on 600A dual IGBT modules (`2MBI600`); (c) `Tektronix TDS 210` digital oscilloscope capture verifying a clean **27.72 kHz** gate-drive square wave; (d) Analog oscilloscope dead-time verification and dual-channel driver testing.

7.2 System-Level Resonant Cabinet Integration, IC Map & Batch PCB Production

Participated in full cabinet electromechanical integration—mounting four parallel `MHS-PCB16` gate drivers onto water-cooled copper busbars and 10-channel cooling manifolds—and operated the temperature-controlled molten solder pot and protective varnish coating for 36-board production batches:

Subsystem PCB	Key Integrated Circuits & Power Devices	Primary Engineering Function & Test Protocol
Frequency & Protection Card (`MHS-PCB19.R1`)	`U3: TL084`, `U18: CD4081`, `U7: CD40106`, `U4: CD4093`, Optocouplers, `7815/7915`	Resonant frequency tuning, dead-time logic conditioning, and isolated `IGBT` / water (`SU`) / thermal (`ISI`) fault interlocks
Display & VCO Control Card	`U2: LM566 (VCO)`, `U6: CA3140`, `U1/U23: TL084`, `U16: LM324`, `U5: LM393`, `U17: CD40106`	Voltage-controlled oscillator frequency generation, current/power feedback scaling, and front-panel operator interface
IGBT Gate Driver & Snubber (`MHS-PCB16`)	`VLA567-11R` Hybrid Gate Driver, `1.5KE` TVS Diodes, Electrolytic Cap Bank, `2MBI600`	Isolated high-current bipolar gate drive (+15V/-8V) and overvoltage transient clamping at 27.72 kHz switching frequency

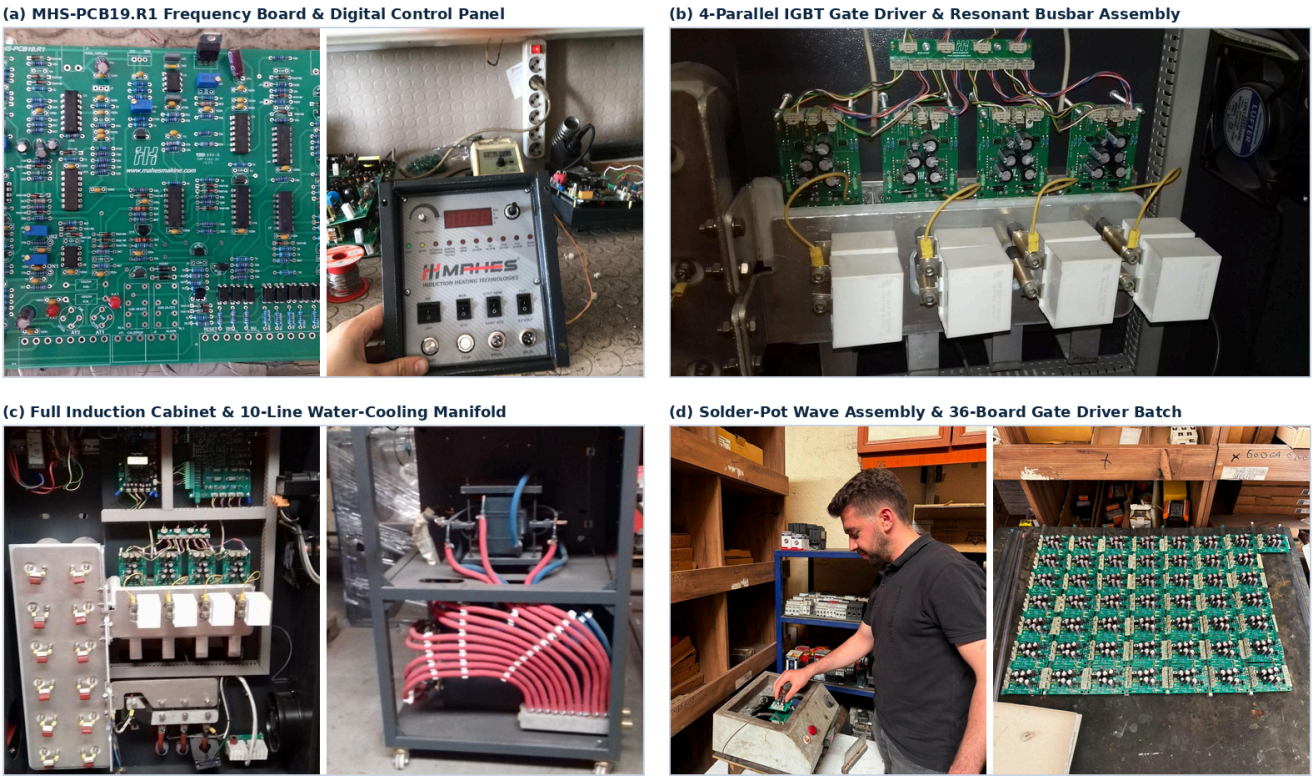


Figure 5: System-level induction heating cabinet integration and batch PCB assembly: (a) `MHS-PCB19.R1` frequency/protection board (top/bottom copper) and front-panel digital controller; (b) Four parallel `MHS-PCB16` gate driver cards mounted atop four high-power IGBT modules and resonant tank capacitors on heavy copper busbars; (c) Full cabinet interior and 10-channel water-cooling manifold with high-frequency output transformer; (d) Molten solder-pot wave soldering operation and a completed batch of 36 `MHS-PCB16` gate driver boards.

Academic Reference (B.Sc. Thesis Committee)	Committee Role	Department & University
Prof. Dr. Uğur Çam	Thesis Advisor (*EED 4894 — Grade: AA*)	Electrical & Electronics Engineering, Dokuz Eylül University
Prof. Dr. Yavuz Şenol	Examination Committee Chairman	Electrical & Electronics Engineering, Dokuz Eylül University
Prof. Dr. Özge Cihanbeğendi & Prof. Dr. Taner Abdullah Oğuzer	Examination Committee Members	Electrical & Electronics Engineering, Dokuz Eylül University